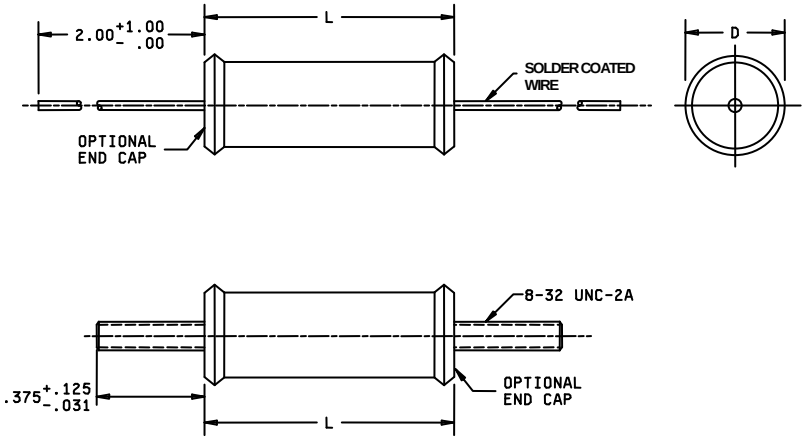
INCH-POUND
MIL-PRF-19978/2E
15 October 2009
SUPERSEDING
MIL-PRF-19978/2D
27 May 1999

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, (HERMETICALLY SEALED IN CERAMIC OR GLASS CASES), STYLE CQ20

This specification sheet is approved for use by all Departments and Agencies of Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19978.



Inches	mm	Inches	mm
.001	0.03	.375	9.53
.004	0.10	.531	13.49
.031	0.79	.625	15.88
.032	0.81	.750	19.05
.040	1.02	1.00	25.4
.125	3.18	2.00	50.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Leads shall be solder coated solid wire, .032 (No. 20 AWG) for cases .531 and .625 in diameter, and .040 (No. 18 AWG) for cases .75 and above in diameter. Tolerance on all lead wire diameters shall be +.004, -.001.
4. Mounting hardware consisting of two lockwashers and four nuts shall be supplied with each capacitor with threaded-stud terminals.
5. These capacitors are not intended to be supported by their leads. These capacitors shall be supported with a supplementary means of mounting, such as tangential retainer or wrap-around band. The supporting device will not be supplied with the capacitor.

FIGURE 1. Style CQ20 capacitors.

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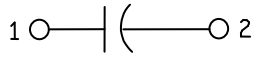
REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Construction: Extended foil or laid-in-tab .

Terminals: See table I.

Circuit diagram: 

Rated voltage: See table I.

Rated temperature: -65°C to +125°C

Capacitance (Cap.) (nom): See table I.

Capacitance tolerance: See table I.

Dissipation factor (DF) (max): 1.0 percent.

Seal: In accordance with method 112 of MIL-STD-202, test condition letter A.

Dielectric withstanding voltage (DWV):

Sleeving: In accordance with MIL-PRF-19978.

Barometric pressure: In accordance with MIL-PRF-19978.

Test points: Circuit diagram 1 - Between terminals and case.

Insulation resistance (IR):

Sleeving - In accordance with MIL-PRF-19978.

Terminal to terminal - See table II.

Vibration, low frequency: Method 201 of MIL-STD-202.

Direction and duration of motion: Four hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt atmosphere (corrosion): Not applicable.

Immersion:

DWV:

Sleeving: In accordance with MIL-PRF-19978.

IR:

Sleeving: In accordance with MIL-PRF-19978.

Solderability: In accordance with MIL-PRF-19978 for wire leads only.

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Terminal strength: In accordance with method 211 of MIL-STD-202.

Wire leads: Test condition letter D.

Threaded stud and nuts: Test condition letters A and E.

Moisture resistance:

DWV:

Sleeving: In accordance with MIL-PRF-19978.

IR:

Sleeving: In accordance with MIL-PRF-19978.

Stability at low and high temperatures:

Low temperature:

Test temperature: -65°C +0°C, -5°C.

Capacitance change (max): -10 percent.

High temperature:

Test temperature: +125°C +0°C, -5°C.

Capacitance change (max): +10 percent.

Life:

Capacitance change (max): ±5 percent.

Resistance to soldering heat: In accordance with MIL-PRF-19978.

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TABLE I. Style CQ20 capacitors.

Part or identifying number (PIN) 1/	DC voltage rating (see figure 2)	Capacitance	Capacitance tolerance	Terminal identification	Case dimensions 2/	
					L ±.062	D ±.031
	Volts	µF			Inches	Inches
CQ20A1KG102-1	1,000	0.0010	J, K	A	1.188	.531
CQ20A1KG152-1	1,000	0.0015	J, K	A	1.188	.531
CQ20A1KG222-1	1,000	0.0022	J, K	A	1.188	.531
CQ20A1KG332-1	1,000	0.0033	J, K	A	1.188	.531
CQ20A1KG472-1	1,000	0.0047	J, K	A	1.188	.531
CQ20A1KG682-1	1,000	0.0068	J, K	A	1.188	.531
CQ20A1KG103-1	1,000	0.010	J, K	A	1.188	.531
CQ20A1KG153-1	1,000	0.015	J, K	A	1.188	.531
CQ20A1KG223-1	1,000	0.022	J, K	A	1.562	.531
CQ20-1KG333-1	1,000	0.033	J, K	A,C	1.562	.625
CQ20-1KG473-1	1,000	0.047	J, K	A,C	1.750	.625
CQ20-1KG683-1	1,000	0.068	J, K	A,C	2.000	.625
CQ20-1KG104-1	1,000	0.10	J, K	A,C	2.250	.625
CQ20-1KG154-1	1,000	0.15	J, K	A,C	2.250	.750
CQ20-1KG224-1	1,000	0.22	J, K	A,C	2.500	.750
CQ20-1KG334-1	1,000	0.33	J, K	A,C	2.750	1.031
CQ20-1KG474-1	1,000	0.47	J, K	A,C	3.000	1.031
CQ20-1KG684-1	1,000	0.68	J, K	A,C	3.250	1.344
CQ20-1KG105-1	1,000	1.0	J, K	A,C	3.750	1.344
CQ20A1KJ102-1	2,000	0.0010	J, K	A	1.188	.531
CQ20A1KJ152-1	2,000	0.0015	J, K	A	1.188	.531
CQ20A1KJ222-1	2,000	0.0022	J, K	A	1.188	.531
CQ20A1KJ332-1	2,000	0.0033	J, K	A	1.188	.531
CQ20A1KJ472-1	2,000	0.0047	J, K	A	1.188	.531
CQ20A1KJ682-1	2,000	0.0068	J, K	A	1.188	.531
CQ20A1KJ103-1	2,000	0.010	J, K	A	1.562	.531
CQ20A1KJ153-1	2,000	0.015	J, K	A	1.562	.531
CQ20-1KJ223-1	2,000	0.022	J, K	A,C	1.562	.625
CQ20-1KJ333-1	2,000	0.033	J, K	A,C	1.750	.625
CQ20-1KJ473-1	2,000	0.047	J, K	A,C	2.000	.625
CQ20-1KJ683-1	2,000	0.068	J, K	A,C	2.000	.750
CQ20-1KJ104-1	2,000	0.10	J, K	A,C	2.250	.875
CQ20-1KJ154-1	2,000	0.15	J, K	A,C	2.500	1.031
CQ20-1KJ224-1	2,000	0.22	J, K	A,C	2.750	1.031
CQ20-1KJ334-1	2,000	0.33	J, K	A,C	2.750	1.344
CQ20-1KJ474-1	2,000	0.47	J, K	A,C	3.750	1.344
CQ20-1KJ684-1	2,000	0.68	J, K	A,C	3.750	1.344
CQ20A1KL102-1	3,000	0.001	J, K	A	1.188	.531
CQ20A1KL152-1	3,000	0.0015	J, K	A	1.188	.531
CQ20A1KL222-1	3,000	0.0022	J, K	A	1.188	.531
CQ20A1KL332-1	3,000	0.0033	J, K	A	1.188	.531
CQ20A1KL472-1	3,000	0.0047	J, K	A	1.562	.531
CQ20A1KL682-1	3,000	0.0068	J, K	A	1.562	.531
CQ20-1KL103-1	3,000	0.010	J, K	A,C	1.562	.625
CQ20-1KL153-1	3,000	0.015	J, K	A,C	1.750	.625
CQ20-1KL223-1	3,000	0.022	J, K	A,C	2.000	.625
CQ20-1KL333-1	3,000	0.033	J, K	A,C	2.000	.750
CQ20-1KL473-1	3,000	0.047	J, K	A,C	2.250	.750
CQ20-1KL683-1	3,000	0.068	J, K	A,C	2.500	.875
CQ20-1KL104-1	3,000	0.10	J, K	A,C	3.250	.875

See footnotes at end of table.

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TABLE I. Style CQ20 capacitors - Continued.

Part or identifying number (PIN) 1/	DC voltage rating (see figure 2)	Capacitance	Capacitance tolerance	Terminal identification	Case dimensions 2/	
					L ±.062	D ±.031
	Volts	µF			Inches	Inches
CQ20-1KL154-1	3,000	0.15	J, K	A,C	3.250	1.031
CQ20-1KL224-1	3,000	0.22	J, K	A,C	3.250	1.344
CQ20-1KL334-1	3,000	0.33	J, K	A,C	3.750	1.344
CQ20A1KN471-1	5,000	0.00047	J, K	A	1.562	.531
CQ20A1KN681-1	5,000	0.00068	J, K	A	1.562	.531
CQ20A1KN102-1	5,000	0.0010	J, K	A	1.562	.531
CQ20A1KN152-1	5,000	0.0015	J, K	A	1.562	.531
CQ20A1KN222-1	5,000	0.0022	J, K	A	1.562	.531
CQ20A1KN332-1	5,000	0.0033	J, K	A	1.562	.531
CQ20-1KN472-1	5,000	0.0047	J, K	A,C	1.562	.625
CQ20-1KN682-1	5,000	0.0068	J, K	A,C	1.750	.625
CQ20-1KN103-1	5,000	0.010	J, K	A,C	2.250	.625
CQ20-1KN153-1	5,000	0.015	J, K	A,C	2.250	.750
CQ20-1KN223-1	5,000	0.022	J, K	A,C	2.500	.875
CQ20-1KN333-1	5,000	0.033	J, K	A,C	3.000	.875
CQ20-1KN473-1	5,000	0.047	J, K	A,C	3.000	1.031
CQ20-1KN683-1	5,000	0.068	J, K	A,C	3.500	1.031
CQ20-1KN104-1	5,000	0.10	J, K	A,C	3.500	1.344
CQ20-1KN154-1	5,000	0.15	J, K	A,C	3.750	1.344
CQ20-1KR471-1	7,500	0.00047	J, K	A,C	1.562	.625
CQ20-1KR681-1	7,500	0.00068	J, K	A,C	1.562	.625
CQ20-1KR102-1	7,500	0.0010	J, K	A,C	1.750	.625
CQ20-1KR152-1	7,500	0.0015	J, K	A,C	1.750	.625
CQ20-1KR222-1	7,500	0.0022	J, K	A,C	1.750	.750
CQ20-1KR332-1	7,500	0.0033	J, K	A,C	2.250	.750
CQ20-1KR472-1	7,500	0.0047	J, K	A,C	2.250	.750
CQ20-1KR682-1	7,500	0.0068	J, K	A,C	2.750	.875
CQ20-1KR103-1	7,500	0.010	J, K	A,C	2.750	.875
CQ20-1KR153-1	7,500	0.015	J, K	A,C	2.750	1.031
CQ20-1KR223-1	7,500	0.022	J, K	A,C	2.750	1.344
CQ20-1KR333-1	7,500	0.033	J, K	A,C	2.750	1.344
CQ20-1KR473-1	7,500	0.047	J, K	A,C	3.500	1.344
CQ20-1KR683-1	7,500	0.068	J, K	A,C	4.250	1.344
CQ20-1KS471-1	10,000	0.00047	J, K	A,C	2.250	.625
CQ20-1KS681-1	10,000	0.00068	J, K	A,C	2.250	.625
CQ20-1KS102-1	10,000	0.0010	J, K	A,C	2.250	.750
CQ20-1KS152-1	10,000	0.0015	J, K	A,C	2.250	.875
CQ20-1KS222-1	10,000	0.0022	J, K	A,C	2.750	.875
CQ20-1KS332-1	10,000	0.0033	J, K	A,C	2.750	.875
CQ20-1KS472-1	10,000	0.0047	J, K	A,C	2.750	1.031
CQ20-1KS682-1	10,000	0.0068	J, K	A,C	2.750	1.344
CQ20-1KS103-1	10,000	0.010	J, K	A,C	2.750	1.344
CQ20-1KS153-1	10,000	0.015	J, K	A,C	3.250	1.344
CQ20-1KS223-1	10,000	0.022	J, K	A,C	3.750	1.344
CQ20-1KS333-1	10,000	0.033	J, K	A,C	4.750	1.344
CQ20-1KS393-1	10,000	0.039	J, K	A,C	5.250	1.344
CQ20-1KT471-1	12,500	0.00047	J, K	A,C	2.750	.750
CQ20-1KT681-1	12,500	0.00068	J, K	A,C	2.750	.750
CQ20-1KT102-1	12,500	0.0010	J, K	A,C	2.750	.875
CQ20-1KT152-1	12,500	0.0015	J, K	A,C	3.250	.875

See footnotes at end of table.

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TABLE I. Style CQ20 capacitors - Continued.

Part or identifying number (PIN) 1/	DC voltage rating (see figure 2)	Capacitance	Capacitance tolerance	Terminal identification	Case dimensions 2/	
					L ±.062	D ±.031
	Volts	µF			Inches	Inches
CQ20-1KT222-1	12,500	0.0022	J, K	A,C	3.250	1.031
CQ20-1KT332-1	12,500	0.0033	J, K	A,C	3.750	1.031
CQ20-1KT472-1	12,500	0.0047	J, K	A,C	3.750	1.031
CQ20-1KT682-1	12,500	0.0068	J, K	A,C	3.250	1.344
CQ20-1KT103-1	12,500	0.010	J, K	A,C	3.750	1.344
CQ20-1KT153-1	12,500	0.015	J, K	A,C	4.750	1.344
CQ20-1KT223-1	12,500	0.022	J, K	A,C	5.250	1.344
CQ20-1KU471-1	15,000	0.00047	J, K	A,C	2.750	.750
CQ20-1KU681-1	15,000	0.00068	J, K	A,C	2.750	.875
CQ20-1KU102-1	15,000	0.0010	J, K	A,C	3.250	.875
CQ20-1KU152-1	15,000	0.0015	J, K	A,C	3.250	.875
CQ20-1KU222-1	15,000	0.0022	J, K	A,C	3.250	1.031
CQ20-1KU332-1	15,000	0.0033	J, K	A,C	3.250	1.344
CQ20-1KU472-1	15,000	0.0047	J, K	A,C	3.250	1.344
CQ20-1KU682-1	15,000	0.0068	J, K	A,C	3.750	1.344
CQ20-1KU103-1	15,000	0.010	J, K	A,C	4.250	1.344
CQ20-1KU153-1	15,000	0.015	J, K	A,C	5.250	1.344

1/ Complete part number shall include symbols to indicate terminal and capacitance tolerances.
2/ See table III for metric equivalents.

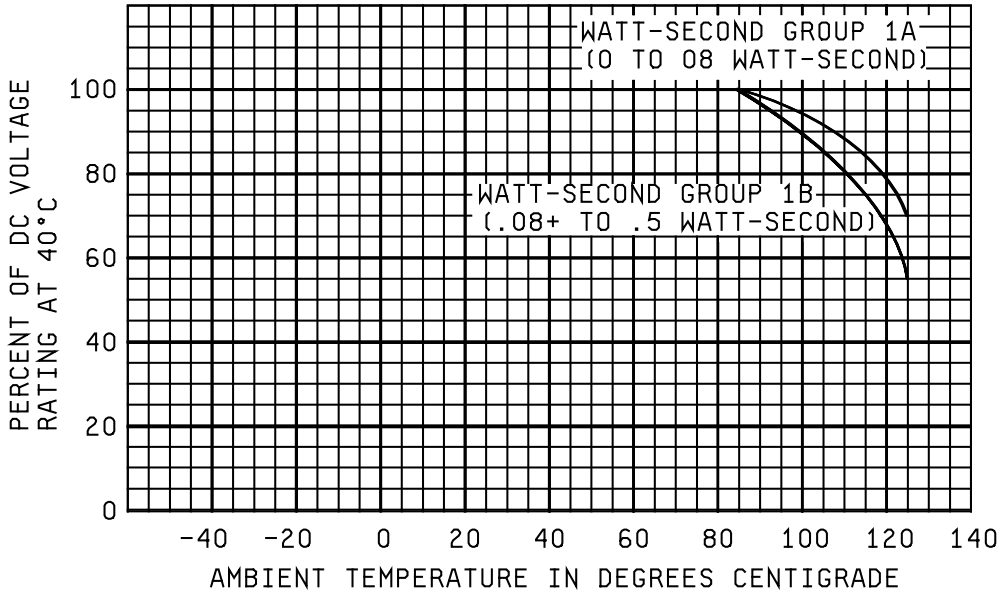


FIGURE 2 .Voltage derating for high ambient test temperature (characteristic K).

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TABLE II. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

TABLE III. Millimeter equivalents of decimal inches.

Inches	mm	Inches	mm
.031	0.79	2.000	50.80
.062	1.57	2.250	57.15
.531	13.49	2.500	63.50
.625	15.88	2.750	69.85
.750	19.05	3.000	75.20
.875	22.23	3.250	82.55
1.031	26.19	3.500	88.90
1.188	30.18	3.750	95.25
1.344	34.14	4.250	107.95
1.562	39.67	4.750	114.30
1.750	44.45	5.250	133.35

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents: In addition to MIL-PRF-19978, this document also references MIL-STD-202.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2009-046)

Review activities:
Navy - MC
Air Force - 19, 99

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